

5. [Maximum mark: 8]

Consider a sequence of ten rectangular picture frames $F_1, F_2, \dots, F_9, F_{10}$.

Picture frame F_1 has width 4 cm and height 5 cm.

The width and height of picture frame F_n , are each increased by 50% to generate the width and height of the next picture frame F_{n+1} , for $n \in \mathbb{Z}^+$, $1 \leq n \leq 9$.

- (a) (i) Show that the area of picture frame F_n is $20\left(\frac{9}{4}\right)^{n-1} \text{ cm}^2$.
- (ii) Hence, find the mean area of the ten picture frames, giving your answer in the form $p\left(\left(\frac{9}{4}\right)^a - 1\right) \text{ cm}^2$, where $p \in \mathbb{Q}^+$, $a \in \mathbb{Z}^+$. [5]
- (b) Find the median area of the ten picture frames, giving your answer in the form $q\left(\frac{9}{4}\right)^4 \text{ cm}^2$, where $q \in \mathbb{Q}^+$. [3]

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